

Star Wars Action Figures Analysis

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Predicting Star Wars Action Figure Pricing

The Star Wars franchise has captivated audiences worldwide for decades, with its impact extending far beyond the big screen. One particularly intriguing aspect of the franchise's legacy is the market for Star Wars action figures. These collectibles vary in value based on factors like rarity, character significance, and popularity driven by media appearances. The problem we address in this project is developing a predictive model to estimate the retail pricing value of Star Wars action figures. This is a significant problem to solve because of the collectibles market's highly dynamic and unpredictable nature, where prices can fluctuate dramatically based on various influences, such as character exposure in new media.

Understanding and predicting these price movements would be incredibly useful for businesses, such as online retailers, auction platforms, and collectible shops, looking to optimize their inventory, pricing strategies, and auction reserves. Collectors can also benefit from data-driven insights that allow them to make informed decisions on which action figures are worth purchasing or holding onto for future gains. Thus, the ultimate goal of this project is to bring a level of predictability and insight to a market traditionally driven by fan enthusiasm and speculation.

Stakeholder Pitch

To gain stakeholder buy-in for this project, I would pitch the problem: "The Star Wars action figure market is a multi-million dollar industry with great volatility. By harnessing data, we can bring a strategic advantage to this market. Our analysis can identify trends and predict which action figures will likely increase in value, allowing businesses to make smarter inventory, marketing, and pricing decisions. For collectors, we provide valuable insights that take the guesswork out of purchasing, helping them build collections that can be appreciated significantly over time. This project isn't just about predicting numbers—it's about transforming how the industry approaches the collectibles market, using advanced data analytics to stay ahead of competitors and capture emerging trends."

Data Sources

The data used in this project was obtained by web scraping several key websites related to the Star Wars franchise and action figures. The primary sources included the Star Wars Databank, which provided detailed character information, and Action Figures 411, a platform offering insights into action figure pricing, release dates, and conditions. These datasets were combined to form a comprehensive base integrating character attributes, media appearances, and historical price data. Additional data was gathered from other sources, such as auction sites, to extend the scope of analysis, including pricing trends and evaluating the impact of new media releases on action figure popularity. Together, these data sources create a rich dataset that allows us to model and predict action figure valuations.

Milestone Summaries

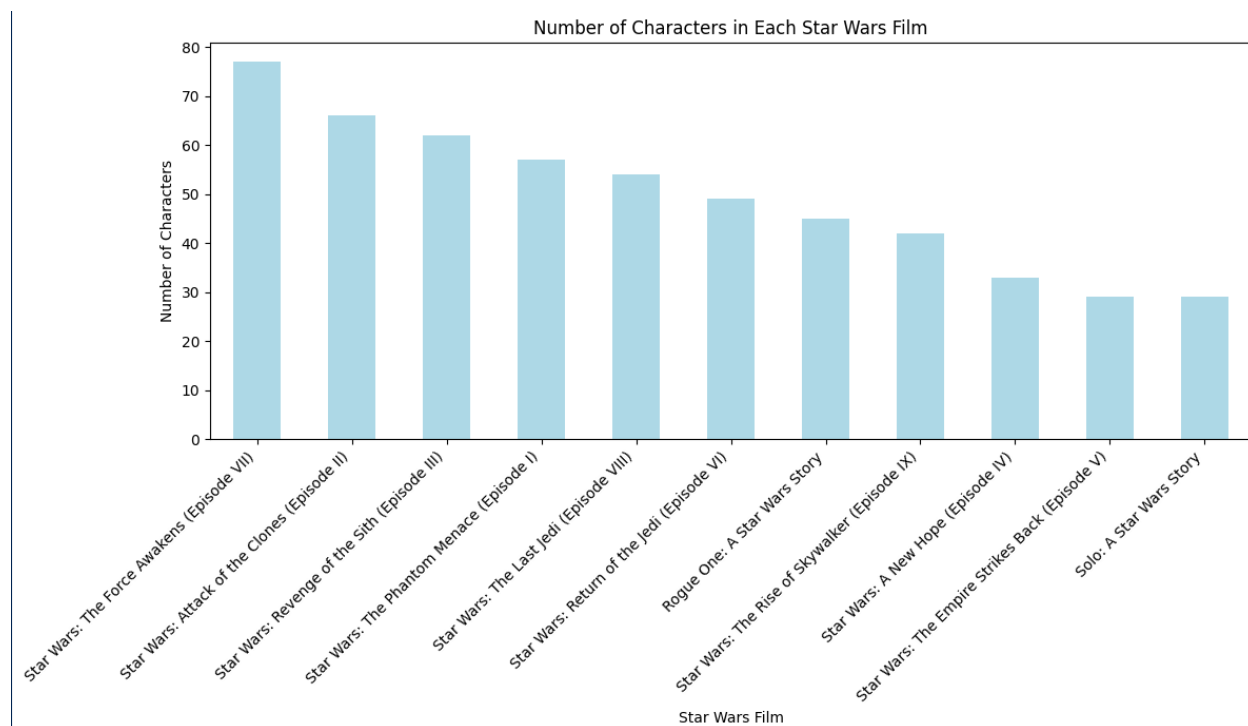
Milestone 1: Exploratory Data Analysis (EDA)

The initial milestone focused on understanding the characteristics of the collected data through Exploratory Data Analysis (EDA). The EDA process involved examining the Star Wars characters and their associated action figures to identify patterns and trends that could impact pricing. Key

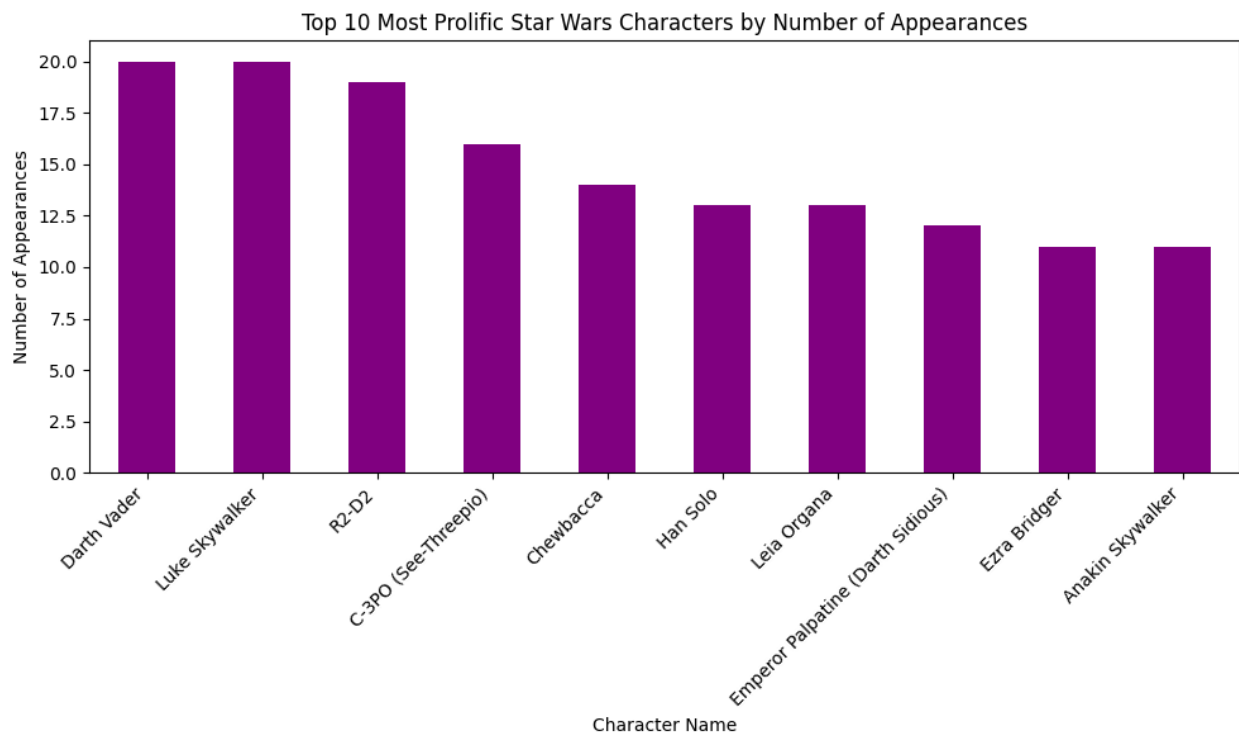
insights included the distribution of character appearances across different media types (films, series, games), the relative popularity of specific characters, and differences in character attributes such as species and gender. These factors were visualized using various graphs, including bar charts showing the number of characters per film (Figure 1) and a bar chart illustrating the top ten most prolific characters based on the number of appearances (Figure 2).

One of the notable findings was the significant variance in the number of action figures produced for characters depending on their role in specific films or series. This insight highlighted the importance of media appearances in influencing action figure production and pricing trends. The EDA phase also revealed missing or inconsistent data, which needed to be addressed during the data preparation phase.

Figure 1:



A bar chart depicting the number of characters per film was included to illustrate which movies had the largest and smallest casts, providing insight into the production focus and potential character popularity. This visualization helped identify which films might have influenced the creation of action figures, thereby impacting collectible trends.

Figure 2:

A bar chart illustrating the ten most prolific characters within the Star Wars universe based on the number of appearances throughout the franchise media. This helps demonstrate the likelihood that a particular character would be known to a consumer, which should increase the demand for action figures for that character.

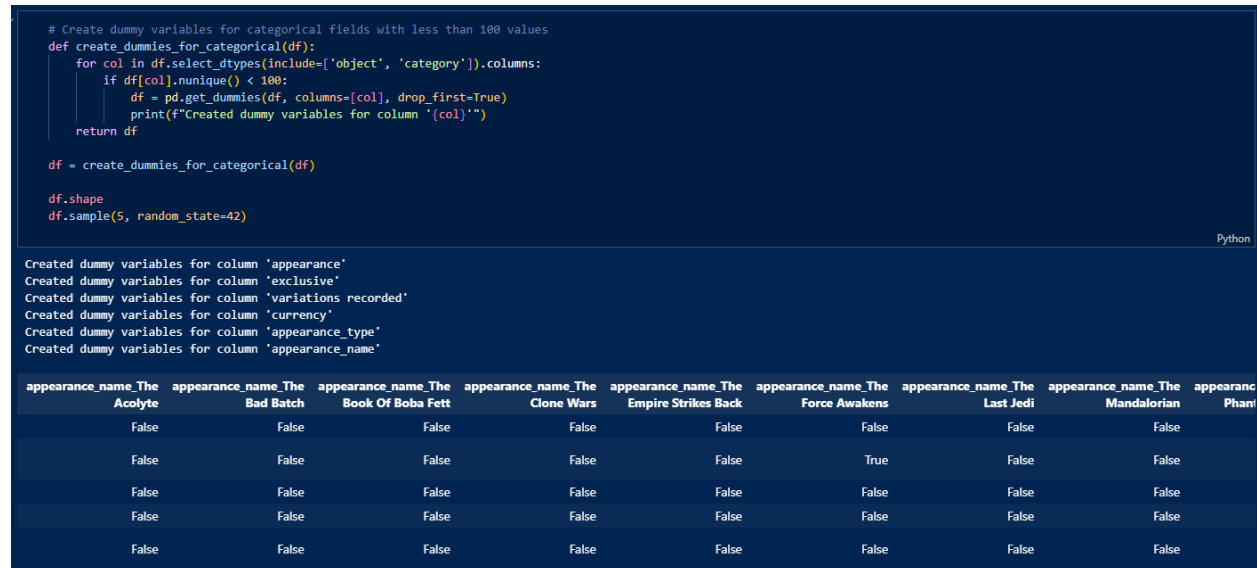
Milestone 2: Data Preparation

In the second milestone, we focused on cleaning and preparing the data for modeling. This involved several key steps: filling in missing values, encoding categorical variables, and creating new features that could improve the predictive power of our model. Missing categorical values, such as the 'exclusive' tag for action figures, were filled with the most common values or set to placeholders like 'Unknown' to ensure data was included in the analysis. For numerical features, median imputation was used to handle missing values, ensuring the central tendency of the data was preserved without the influence of outliers.

Feature engineering played a crucial role in this phase. We created dummy variables for categorical features with a limited number of distinct values, allowing models to effectively use these variables without assuming any false order. High-cardinality features, like character names, were label-encoded to maintain the uniqueness of each character while avoiding the dimensionality problems of one-hot encoding. We also split features such as 'appearance year' into 'start' and 'end' years to capture better when a character was active in the Star Wars media landscape, which could influence the action figure's value.

The final dataset was further transformed to ensure all features were either numeric or properly encoded, ready for input into machine learning models. Outliers, particularly in the MSRP (manufacturer's suggested retail price) feature, were capped at the 90th percentile to mitigate their influence on model training.

Figure 3:



An example showcasing feature encoding methods, such as label encoding vs. dummy variables, was included to demonstrate how categorical data was transformed into numeric form. This visualization is intended to clarify the approach taken to prepare features for modeling, ensuring the model could effectively utilize categorical data.

Milestone 3: Model Building and Evaluation

The third milestone involved building and evaluating different machine-learning models to predict action figure prices. We started with a simple Linear Regression model to establish a baseline, followed by more sophisticated models like Gradient Boosting, CatBoost, LightGBM, and XGBoost. Each model was evaluated using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared to determine their predictive accuracy.

Hyperparameter tuning was performed using GridSearchCV to optimize each model's parameters and improve performance. Ensemble methods, such as a Voting Regressor and a Stacking Regressor, were also explored to combine the strengths of individual models and achieve more robust predictions. The Stacking Regressor, which leveraged predictions from multiple models to train a meta-model, demonstrated the best overall performance, suggesting that combining multiple approaches could capture complex relationships within the data more effectively.

Visualization of the model performance highlighted the significant improvement achieved through ensemble methods compared to individual models (Table 1). The best-performing models provided

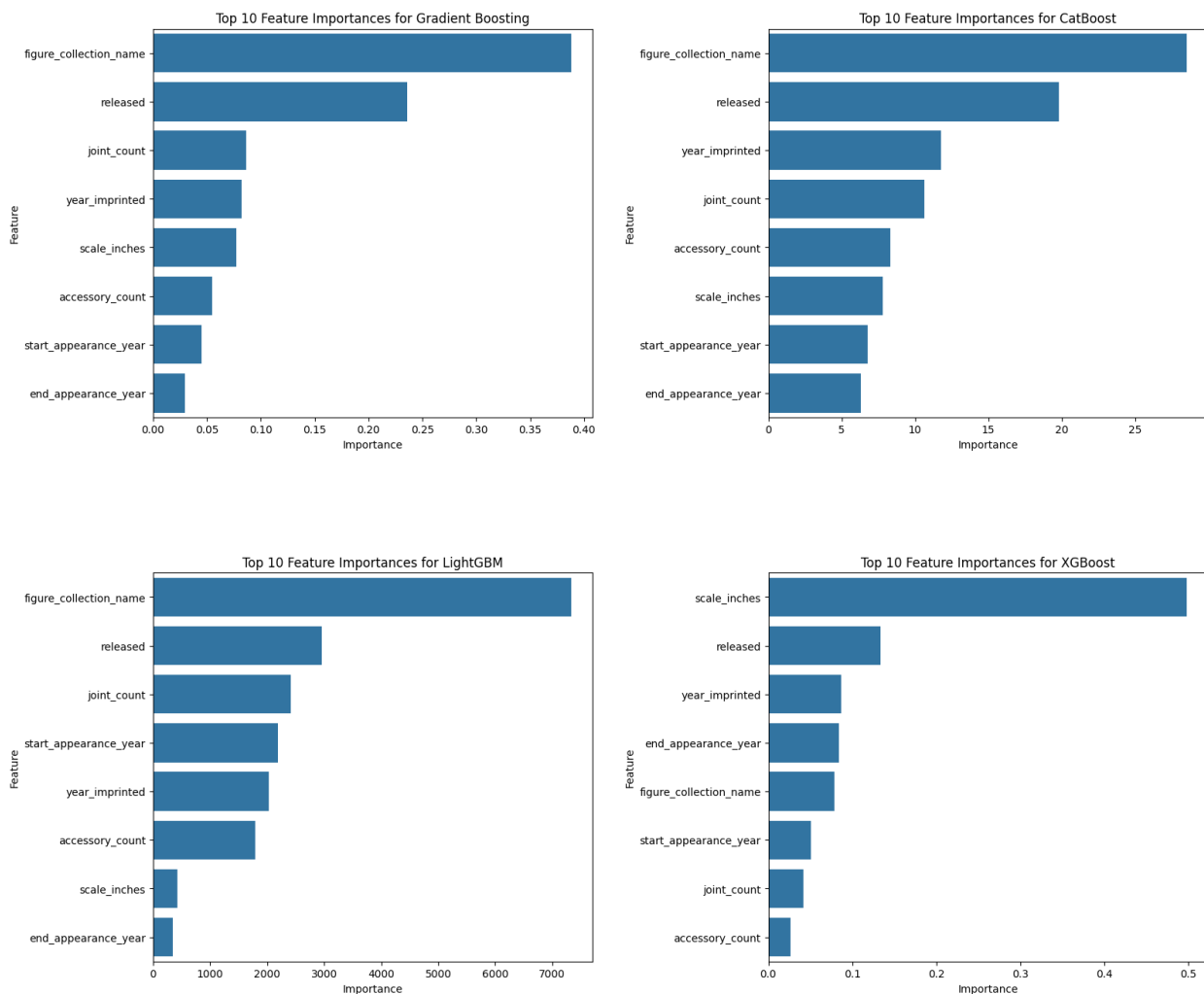
valuable insights into the factors influencing action figure prices, such as character popularity, media appearances, and production details.

Table 1:

	Model	MAE	RMSE	R2	Adjusted R2	MEDAE	MAPE	GridSearch_Time (Seconds)	Model_Train_Predict_Time (Seconds)
0	Gradient Boosting	2.352865	4.739847	0.722329	0.667091	0.430193	7.171683e+13	32.857114	1.007787
1	CatBoost	3.040381	5.042421	0.685746	0.623231	1.375174	1.000074e+14	40.995072	5.011646
2	Linear Regression	5.854970	8.077049	0.193679	0.033276	4.052271	1.015400e+14	NaN	NaN
3	LightGBM	2.862887	4.817707	0.713131	0.656064	1.405696	9.577579e+13	20.360029	0.568775
4	XGBoost	2.395051	4.826912	0.712034	0.654748	0.454591	8.110110e+13	9.742689	1.007787
5	Ensemble	3.050841	4.846657	0.709673	0.651918	1.712302	9.002821e+13	NaN	10.388825
6	Stacking	2.594516	4.595637	0.738968	0.687040	1.058753	8.593065e+13	NaN	21.991027

A comparison table of model performance metrics, including Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), was included to show the relative accuracy of each model. This helped stakeholders quickly see which models performed well and required improvement, providing a basis for selecting the best approach.

Figure 4:



A feature importance plot from the best-performing model was included to illustrate which features had the most significant impact on action figure pricing. This visualization provided insight into the factors driving value, such as media appearances or specific character attributes, and helped guide data collection and feature engineering.

Milestone Summary Conclusion

In conclusion, the milestones allowed us to move from data exploration through data preparation to building robust predictive models. Each step was essential in refining our approach and improving the accuracy of the price predictions, ultimately providing meaningful insights for stakeholders in the Star Wars action figure market.

Project Conclusion

The analysis and model-building process revealed several important insights into the Star Wars action figure market dynamics. Our models identified key value drivers, including media appearances, character popularity, and specific attributes such as rarity or exclusivity. Ensemble methods, particularly the Stacking Regressor, demonstrated that combining different model approaches leads to better predictive accuracy and robustness. This suggests that the market for action figures is influenced by a combination of factors that interact in complex ways, making it well-suited to more advanced machine learning techniques.

While the models show promise in predicting action figure prices, they are not yet ready for deployment in a real-world scenario. There are still limitations related to data availability, particularly concerning pricing history and market trends. Additionally, more granular data on collector behavior, production volumes, and the condition of figures (e.g., mint vs. used) could significantly enhance the model's predictive power. Thus, additional data collection and refinement are needed before the model can be confidently deployed for commercial use.

Recommendations

1. **Data Enrichment:** To improve the accuracy of the models, future work should focus on enriching the dataset. This includes gathering more detailed data on pricing trends over time, production numbers, and information on the condition of action figures. Such data could help capture the nuances of the market and lead to better predictions.
2. **Model Refinement:** The current model could be refined further through targeted hyperparameter tuning and experimentation with additional ensemble approaches. Exploring other machine learning models like neural networks could be worthwhile, particularly if more data becomes available.
3. **Real-Time Data Integration:** Incorporating real-time data, such as new film or series releases and social media buzz, could improve the model's responsiveness to market changes. This would be particularly useful for predicting sudden price spikes linked to media exposure.
4. **Stakeholder Collaboration:** Collaborating with industry stakeholders, such as collectible shops and auction platforms, could help obtain additional data while aligning the model

with business needs. This collaboration could also facilitate early adoption and testing of the model in real-world settings.

Challenges and Future Opportunities

One of the main challenges faced during the project was the availability of comprehensive data. Although web scraping provided a solid foundation, data on historical prices, production volumes, and collector demographics was limited. These gaps need to be addressed to make the model more accurate and reliable. Additionally, market factors such as economic downturns or changes in consumer preferences could introduce unpredictability that our current model is not equipped to handle.

There are also exciting opportunities for further exploration. For instance, using computer vision to analyze images of action figures could provide a new set of features related to figure condition, packaging, and uniqueness. Additionally, leveraging natural language processing (NLP) to analyze fan forums and social media could help gauge collector sentiment, providing another layer of insight into demand drivers. Finally, expanding the model to include other collectibles beyond Star Wars action figures could open new avenues for application in broader collectible markets.

Changes Made Since Milestone 3

Given feedback on the prior milestones, I have not modified any of my earlier analyses for this milestone and instead focused on documenting the methodology and outcomes in this paper. I am open to feedback on additional opportunities I should have addressed, if any.